From methodology to technique

Work documents can be accessed at

https://github.com/prei007/wuhan_workshops/tree/main/theory

1

Workshop thesis

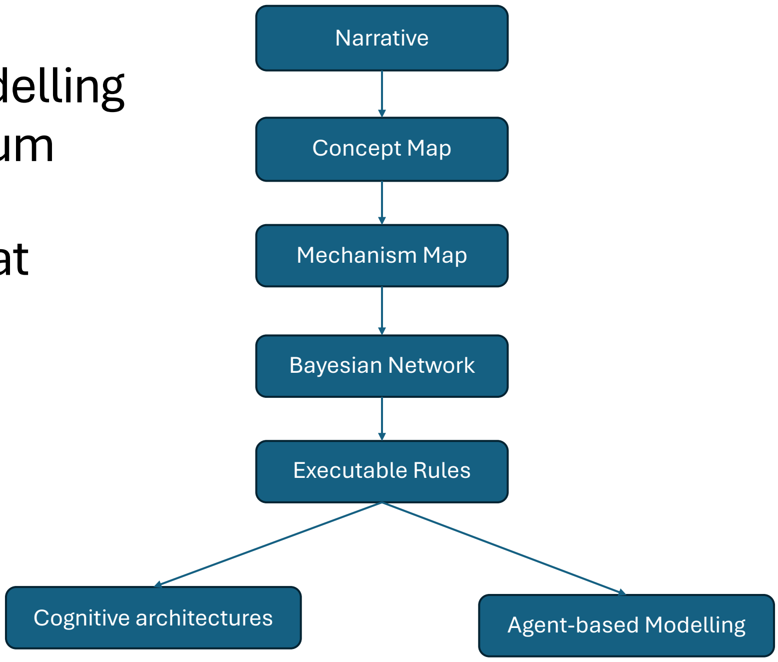
Scientific understanding grows through increasingly powerful external representations (of theory):

- Narratives
- Concept maps
- Diagrams
- Physical models
- Mathematical models
- Computer simulations

Which representation(s) do you currently use for articulating theory in your research?

2

The modelling continuum we'll be looking at



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Step 1: From narrative to concept map

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Narrative theory core (running example)

Many secondary school students use AI writing assistants to revise their essays. Yet AI feedback does not influence revision in a uniform way. Rather, the educational effects of AI depend on how students engage with the feedback. Some students critically evaluate AI suggestions, compare them with their own intentions, and use them to reorganise and strengthen their arguments. Others accept suggestions uncritically or restrict their revisions to surface-level corrections. The central theoretical proposition is that AI feedback does not directly improve writing quality. Instead, it changes the processes through which students evaluate, regulate, and revise their work. Differences in these underlying processes generate differences in the quality of revision and, ultimately, in learning outcomes.

Which nouns and verbs are relevant for theory development?

- Noun-> blue
- Verb -> red

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Concept mapping with Mermaid

https://mermaid.live

```
flowchart TD

Student[Student]
AI[AI Feedback]
Revision[Revision Process]
Essay[Essay]
Quality[Writing Quality]

Student --> Revision
AI --> Revision
Revision --> Essay
Essay --> Quality
```

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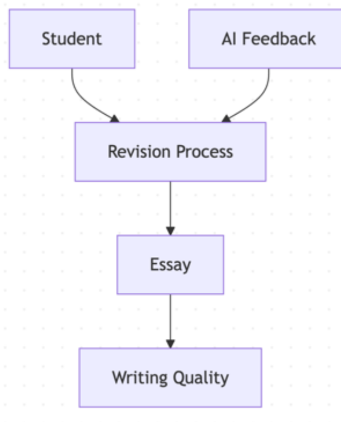
Concept maps organise knowledge

What can a concept map help us do?

What can't it do?



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Build your own concept map (10 minutes)

For this:

[Narrative01.md]

Many secondary school students use AI writing assistants to revise their essays. Yet AI feedback does not influence revision in a uniform way. Rather, the educational effects of AI depend on how students engage with the feedback. Some students critically evaluate AI suggestions, compare them with their own intentions, and use them to reorganise and strengthen their arguments. Others accept suggestions uncritically or restrict their revisions to surface-level corrections. The central theoretical proposition is that AI feedback does not directly improve writing quality. Instead, it changes the processes through which students evaluate, regulate, and revise their work. Differences in these underlying processes generate differences in the quality of revision and, ultimately, in learning outcomes.

Or: for your own research.

Template:

[ConceptMapTemplate.txt]

```
flowchart TD

A[Concept]
B[Concept]
C[Concept]
D[Concept]
E[Concept]

A --> B
A --> C
B --> D
C --> D
D --> E
```

Go to https://mermaid.live

Or draw by hand!

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Compare and discuss

Share in a pair and discuss:

- Which concepts seem missing?
- Which arrows feel vague?

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Step 2: From concept map to mechanism map

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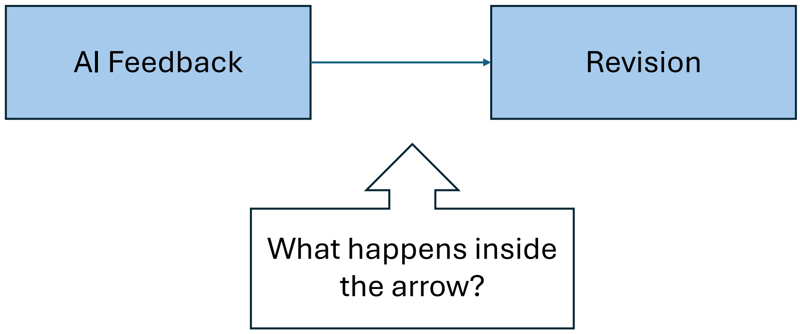
What does happen?



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The mechanism grammar

Four types of "words":

- Context
- Entities
- States
- Activities

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Four questions

Question	Grammar
Where?	Context
Who?	Entity
What changes?	State
How?	Activity

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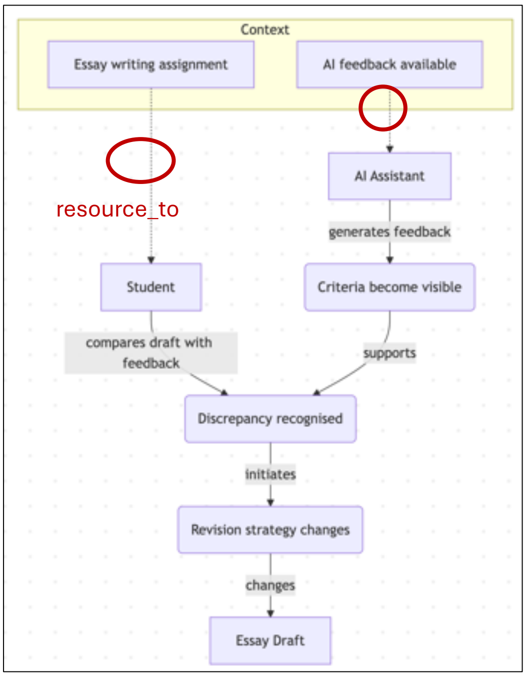
```
flowchart TD

subgraph Context
C1[Essay writing assignment]
C2[AI feedback available]
end

Student[Student]
AI[AI Assistant]
Essay[Essay Draft]

Visible(Criteria become visible)
Gap(Discrepancy recognised)
Revision(Revision strategy changes)

AI -->|generates feedback| Visible
Student -->|compares draft with feedback| Gap
Visible -->|supports| Gap
Gap -->|initiates| Revision
Revision -->|changes| Essay

C1 -.-> Student
C2 -.-> AI
```

[MechanisMap01.txt]



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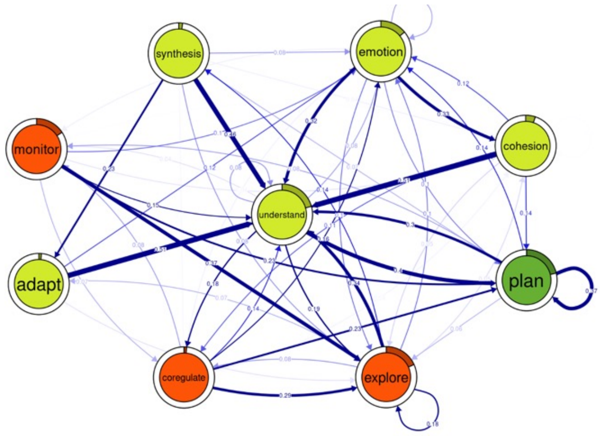

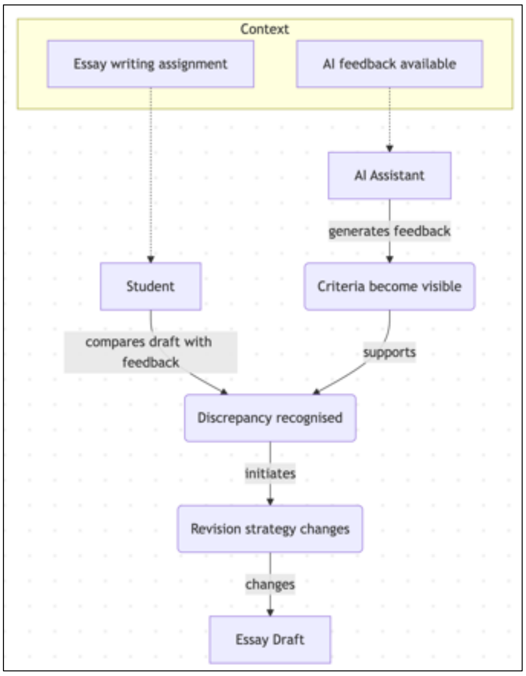

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Theory comes later (only this time)

- Could "Discrepancy recognised" be called "Monitoring:?
- Is "Revision strategy changes" a form of "Strategic regulation"?

High-level theoretical constructs seem to compress recurring mechanisms.

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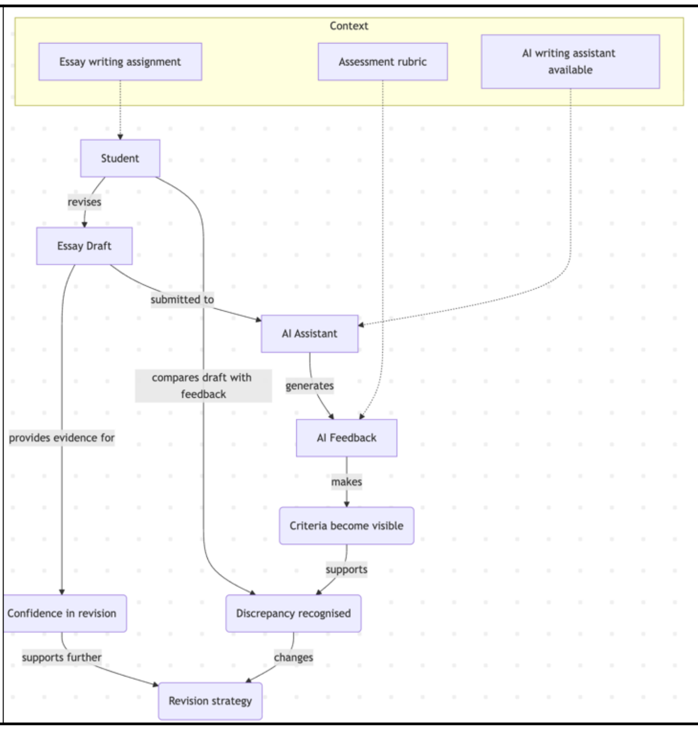
```
flowchart TD

subgraph Context
C1[Essay writing assignment]
C2[Assessment rubric]
C3[AI writing assistant available]
end

Student[Student]
AI[AI Assistant]
Draft[Essay Draft]
Feedback[AI Feedback]

Visible(Criteria become visible)
Gap(Discrepancy recognised)
Revision(Revision strategy)
Confidence(Confidence in revision)

AI -->|generates| Feedback
Draft -->|submitted to| AI

Feedback -->|makes| Visible

Student -->|compares draft with feedback| Gap
Visible -->|supports| Gap

Gap -->|changes| Revision

Student -->|revises| Draft

Draft -->|provides evidence for| Confidence

Confidence -->|supports further| Revision

C1 -.-> Student
C2 -.-> Feedback
C3 -.-> AI
```

[MechanismMap02.txt]



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Activity: Draw your own mechanism map

Use Mermaid or draw by hand

Rules:

- Every arrow (link) must become
  - an activity or
  - an enablement relation (Context factor to entity, use dashed line)
- Each node is a context, entity, or state.

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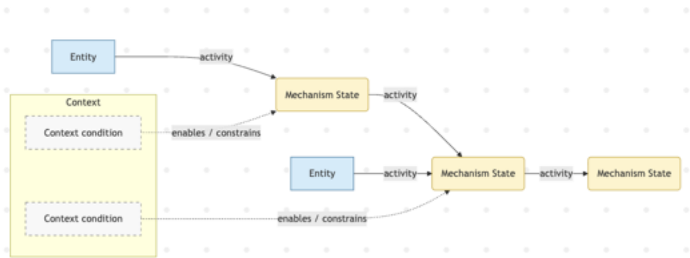
```
flowchart LR

subgraph CONTEXT["Context"]
C1[Context condition]
C2[Context condition]
end

E1[Entity]
E2[Entity]

S1(Mechanism State)
S2(Mechanism State)
S3(Mechanism State)

E1 -->|activity| S1
S1 -->|activity| S2
E2 -->|activity| S2
S2 -->|activity| S3

C1 -.->|enables / constrains| S1
C2 -.->|enables / constrains| S2

classDef context fill:#F8F8F8,stroke:#777,stroke-dasharray:5 5;
classDef entity fill:#D6EAF8,stroke:#1F618D;
classDef state fill:#FCF3CF,stroke:#B7950B;

class C1,C2 context;
class E1,E2 entity;
class S1,S2,S3 state;
```



Mechanism_template-v1.txt

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In pairs or groups: Compare your mechanism maps

- Are the well-formed?
- Which activity is most important?
- Did your explanation improve?

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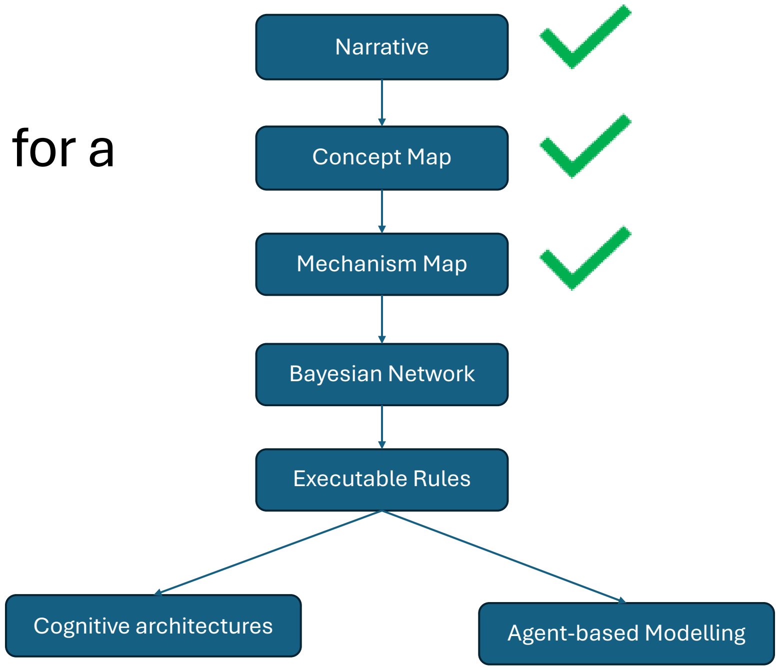
It's time for a break



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Towards counterfactual reasoning: From mechanisms to computational models

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How can we account for differences between students?

Solution: Give students (or any entity, for that matter) attributes

	Attributes	States
Persistence	Relatively stable during the episode	Change during the episode
Examples	Intrinsic motivation, prior knowledge, epistemic beliefs, AI trust disposition	Confusion, discrepancy recognised, monitoring active, confidence, shared understanding
Role	Modulate how the mechanism operates	Constitute the mechanism as it unfolds

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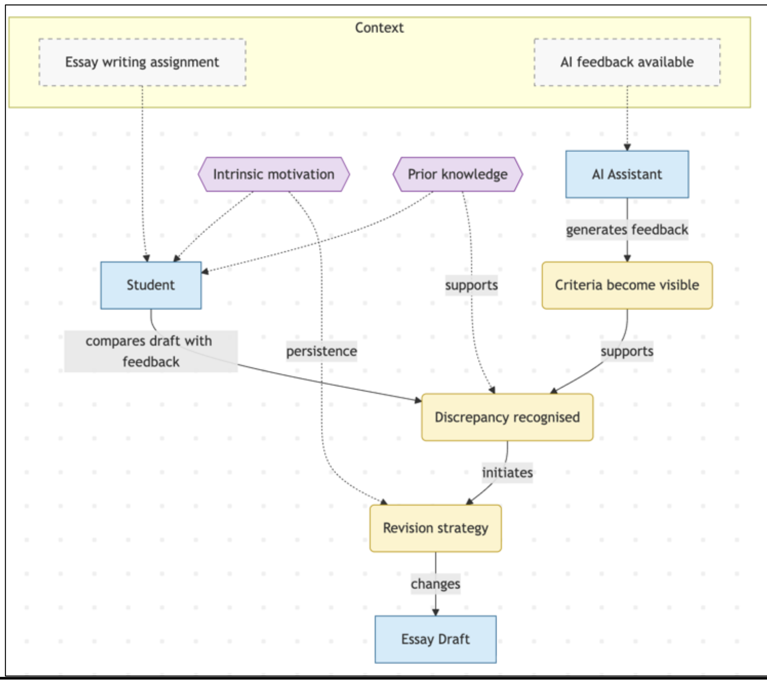
```
flowchart TD

subgraph Context
C1[Essay writing assignment]
C2[AI feedback available]
end

Student[Student]
AI[AI Assistant]
Draft[Essay Draft]

Motivation{{Intrinsic motivation}}
PriorKnowledge{{Prior knowledge}}

Visible(Criteria become visible)
Gap(Discrepancy recognised)
Revision(Revision strategy)

AI -->|generates feedback| Visible
Student -->|compares draft with feedback| Gap
Visible -->|supports| Gap
Gap -->|initiates| Revision
Revision -->|changes| Draft

Motivation -.-> Student
PriorKnowledge -.-> Student

Motivation -.->|persistence| Revision
PriorKnowledge -.->|supports| Gap

C1 -.-> Student
C2 -.-> AI

%% Styles
classDef context fill:#F8F8F8,stroke:#888,stroke-dasharray: 5 5;
classDef entity fill:#D6EAF8,stroke:#1F618D;
classDef attribute fill:#E8DAEF,stroke:#7D3C98;
classDef state fill:#FCF3CF,stroke:#B7950B;

class C1,C2 context;
class Student,AI,Draft entity;
class Motivation,PriorKnowledge attribute;
class Visible,Gap,Revision state;
```

[MechanismMap03.txt]



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Taking stock: node/object classification

Type	Example	Role
Context	Essay assignment	External condition
Entity	Student	Performs activities
Attribute	Intrinsic motivation	Disposition of an entity
Mechanism state	Criteria visible	Changes during execution
Evidence	Essay quality	Observable outcome

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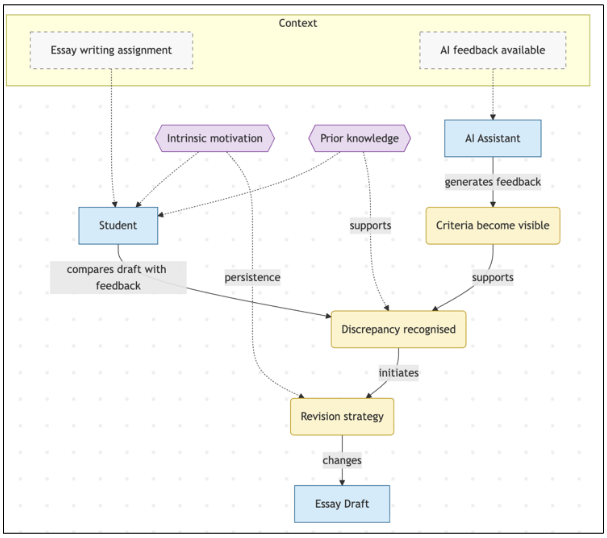
Rich mechanisms are difficult to reason with



Suppose we change one thing:

- Intrinsic motivation becomes low.

Can you mentally work out all consequences?

Probably not; that's why we need computational representations.

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Reducing information richness to employ mathematical modelling

The mechanism map contains:
- Context
- Entities
- Attributes
- Activities
- States

Our next modelling technique (DAG) contains:
- Conditions
- Mechanism states
- Evidence

We loose
- entities and
- activities

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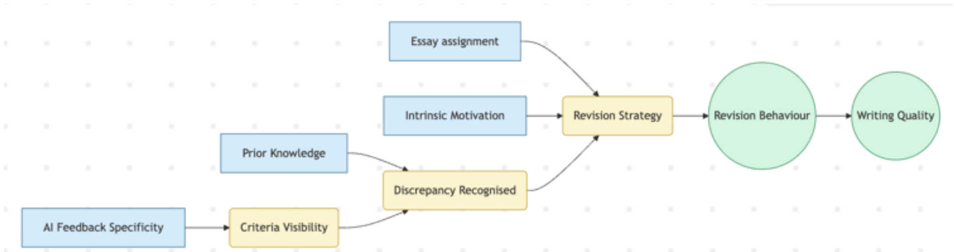
The DAG* visual grammar

Shape	**Meaning**
Rectangle	Conditions (contexts + attributes)
Rounded rectangle	Mechanism states
Circle	Observable evidence



*) Directed Acyclical Graph

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```
flowchart LR

%% Conditions
Context[Essay assignment]
Feedback[AI Feedback Specificity]
Motivation[Intrinsic Motivation]
Knowledge[Prior Knowledge]

%% Hidden mechanism states
Visible(Criteria Visibility)
Gap(Discrepancy Recognised)
Revision(Revision Strategy)

%% Evidence
Behaviour((Revision Behaviour))
Quality((Writing Quality))

Context --> Revision
Feedback --> Visible
Motivation --> Revision
Knowledge --> Gap

Visible --> Gap
Gap --> Revision

Revision --> Behaviour
Behaviour --> Quality

classDef condition fill:#D6EAF8,stroke:#1F618D;
classDef mechanism fill:#FCF3CF,stroke:#B7950B;
classDef evidence fill:#D5F5E3,stroke:#1E8449;

class Context,Feedback,Motivation,Knowledge condition;
class Visible,Gap,Revision mechanism;
class Behaviour,Quality evidence;
```

DAG01

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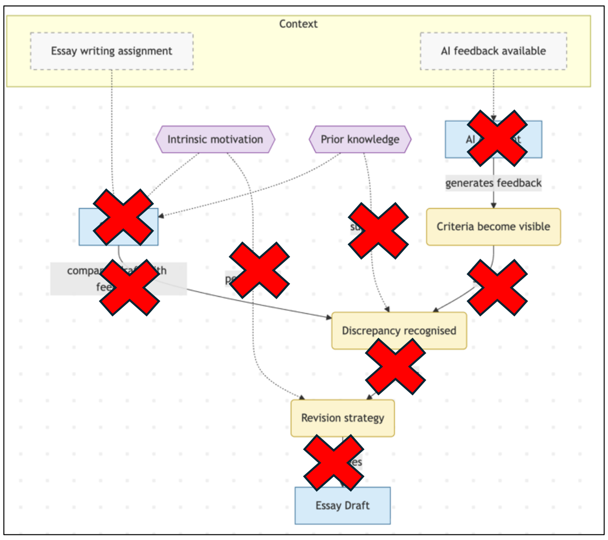

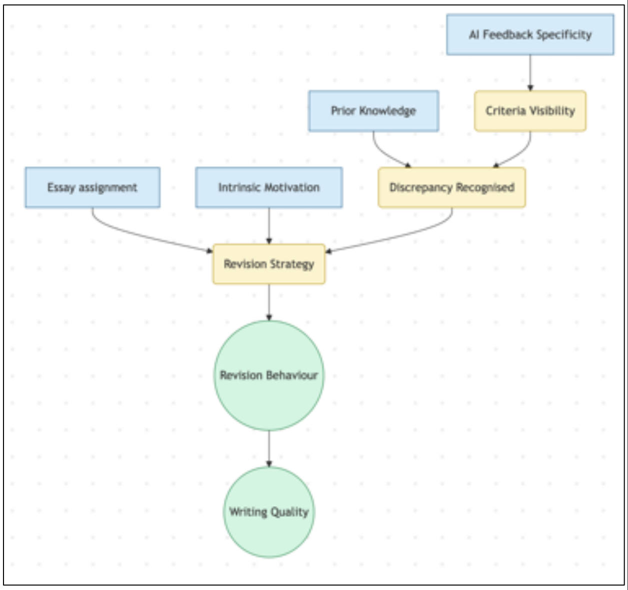

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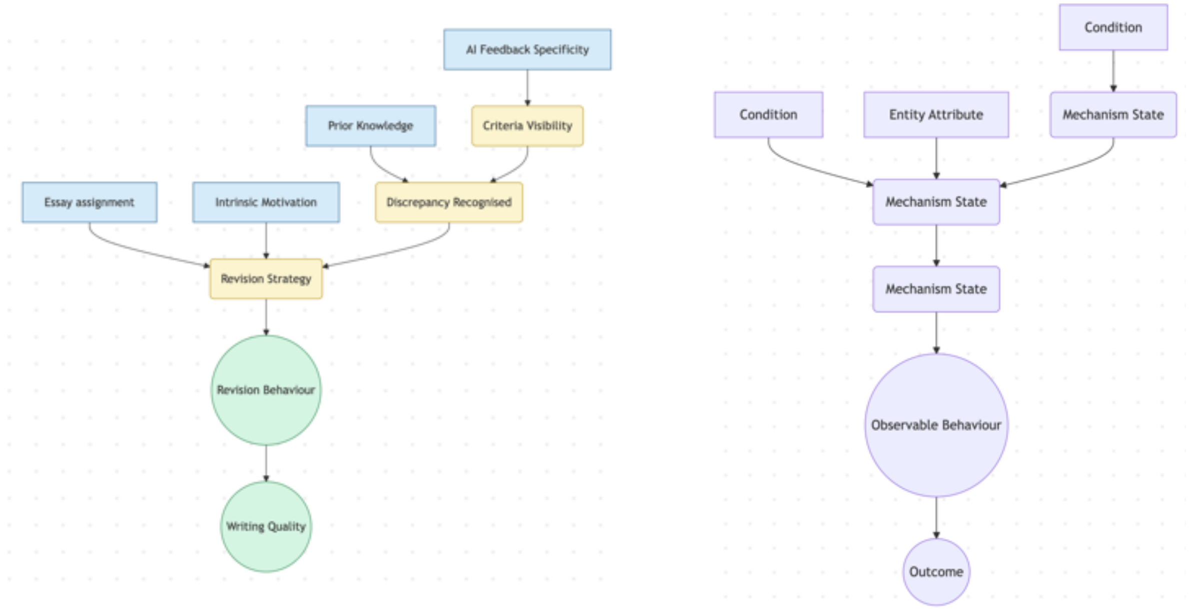

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The representations so far

Representation	Main question	Primary objects
Concept Map	What concepts matter?	Concepts
Mechanism Map	What happens?	Contexts, Entities, Attributes, States, Activities
DAG	What (causally) influences what?	Conditions, Mechanism States, Evidence

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Tool support for asking what-if questions: Bayesian Networks

Step 1: Define states

Step 2: Specify conditional probabilities

Defining states

Principles for a "good" state:

1. States should distinguish meaningfully different situations
2. States should come from theory
3. States should matter for decisions
4. States should support counterfactual reasoning

Trust = {low, medium, high}

Trust = {Distrust, Calibrated trust, Over-trust}

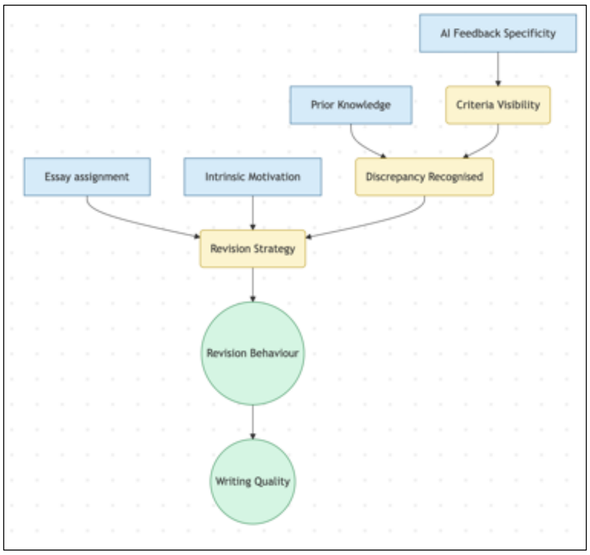
Activity

In pairs:

1. **Pick a node**
2. **Propose 3-5 possible states**

Principles for a "good" state:

1. States should distinguish meaningfully different situations
2. States should come from theory
3. States should matter for decisions
4. States should support counterfactual reasoning



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Three kinds of states

Variable type	Example	Good states
Dispositional (attributes)	Intrinsic motivation	Avoidance • Compliance • Mastery-oriented
Mechanism	Revision strategy	Surface editing • Local restructuring • Global conceptual revision
Observable	Writing quality	Fragmented argument • Coherent argument • Persuasive argument

States for mechanisms are particularly relevant: Every state should correspond to a recognisably different mode of operation of the mechanism.

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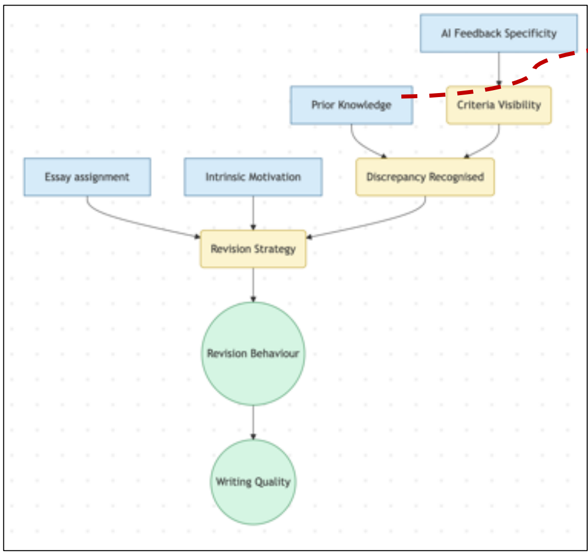
Step 2: Specify (conditional) probabilities



Prior (Writing) Knowledge		
Fragile	Procedural	Strategic
p = .30	p = .50	p = .20

Prior Research

Expert (e.g., teacher) estimation

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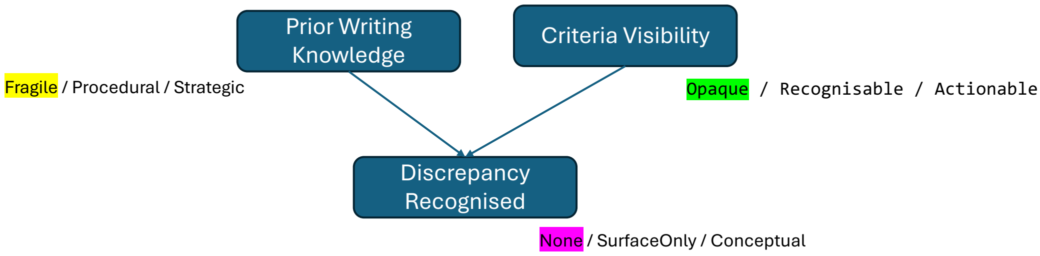
Step 2: Specify (conditional) probabilities



PriorWritingKnowledge	CriteriaVisibility	DiscrepancyRecognised = None	DiscrepancyRecognised = SurfaceOnly	DiscrepancyRecognised = Conceptual
Fragile	Opaque	0.7	0.25	0.05
Fragile	Recognisable	0.45	0.45	0.1
Fragile	Actionable	0.2	0.55	0.25
Procedural	Opaque	0.45	0.45	0.1
Procedural	Recognisable	0.2	0.55	0.25
Procedural	Actionable	0.1	0.45	0.45
Strategic	Opaque	0.2	0.55	0.25
Strategic	Recognisable	0.1	0.45	0.45
Strategic	Actionable	0.05	0.25	0.7

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The representations so far

Representation	**Main question**	**Primary objects**
Concept Map	What concepts matter?	Concepts
Mechanism Map	What happens?	Contexts, Entities, Attributes, States, Activities
DAG	What (causally) influences what?	Conditions, Mechanism States, Evidence
Bayesian Network	What would happen if…?	Conditions, Mechanism States, Evidence

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Mechanisms can be modelled with rules

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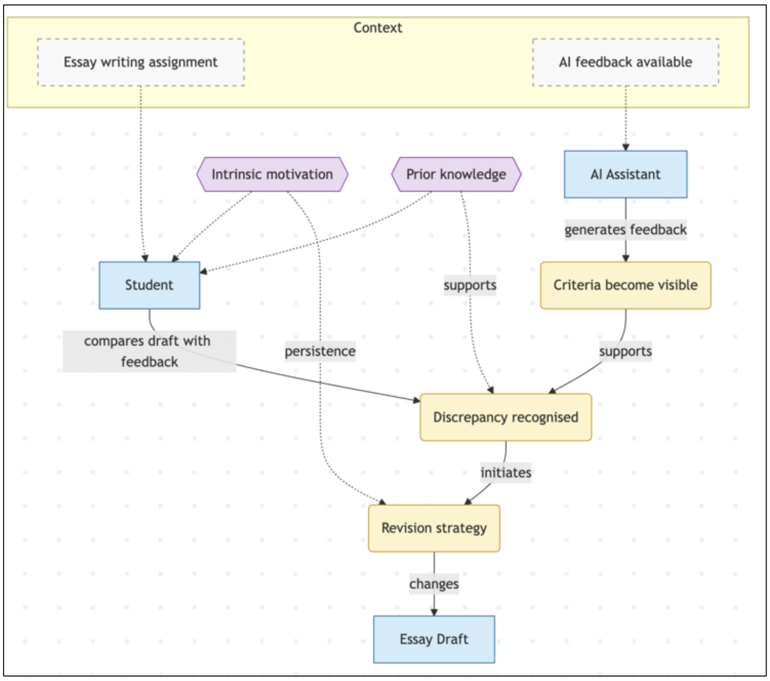

The arrows are still abstractions

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IF...THEN Rules

```
IF discrepancy is recognised
AND intrinsic motivation is mastery-oriented

THEN perform conceptual revision.
```

What kind of scientific object is this?

It is an executable object that combines context, control, action, and outcome in a generative manner.

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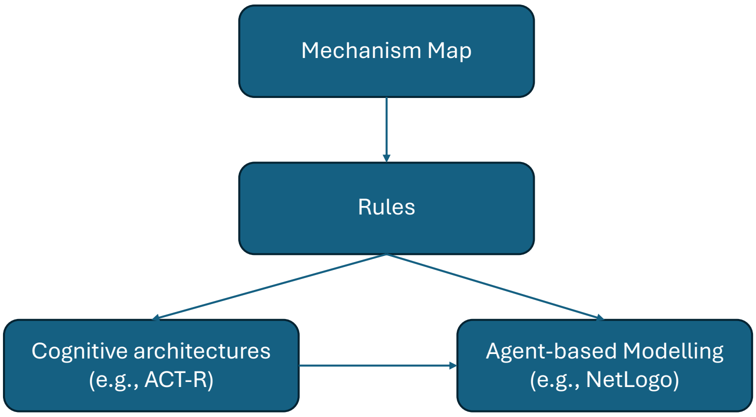
Two rule-based modelling traditions



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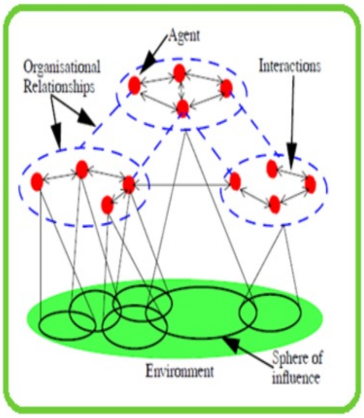
Agent-based models/systems



Kharb & Sukic, TEM Journal – Volume 4 / Number 3 / 2015. www.temjournal.com

Agents interacting in an environment following (simple) rules.

Agents have
- Attributes
- Beliefs
- Simple decision rules

```
IF confidence is low
THEN ask AI.
```

Interesting (system) behaviour emerges from many agents interacting repeatedly.

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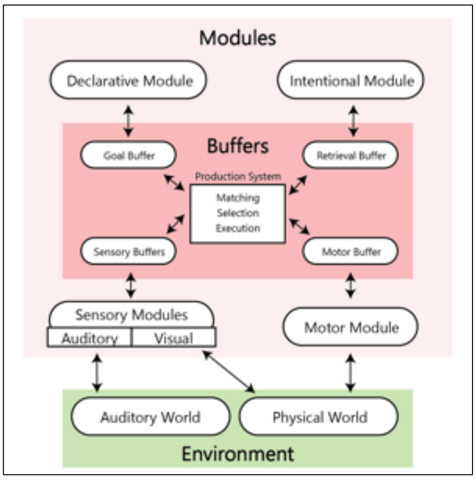
Cognitive architectures: ACT-R



Can help answer the question:
What happens inside one learner?

- Complex reasoning and decision making
- Takes into account brain architecture

```
IF
goal = revise essay
AND discrepancy recognised
THEN
retrieve revision strategy.
```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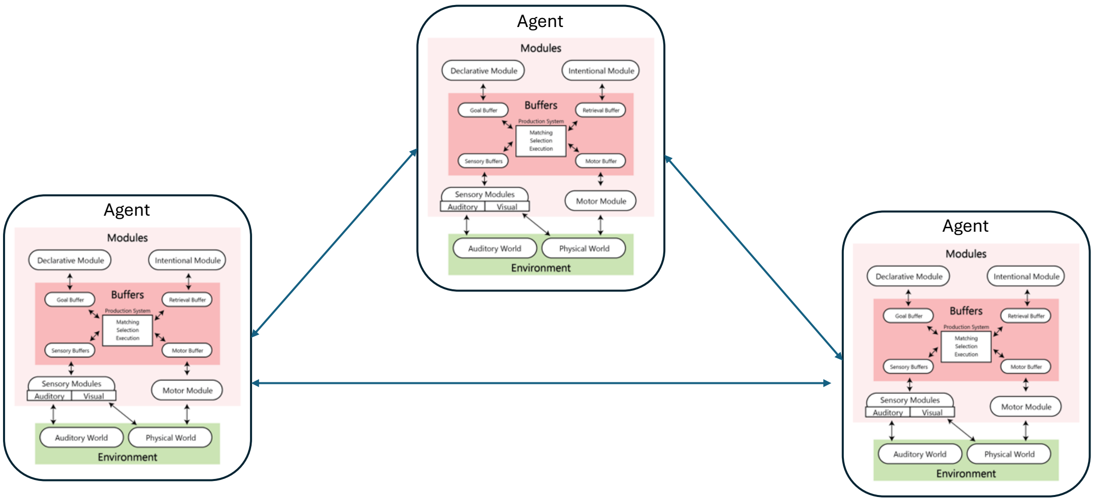
Combining cognitive and ABM approaches



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Mechanistic view, different representations

Representation	**Question**
Mechanism Map	What happens?
DAG	What causes what?
Bayesian Network	What if…?
Agent-Based Model	What emerges from interacting agents?
ACT-R	How does one cognitive agent generate behaviour?

Generative representations

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Different questions, different representations

Research question	**Representation**
What concepts matter?	Concept Map
How is the phenomenon generated?	Mechanism Map
What would happen if…?	Bayesian Network
How do many people and technologies interact over time?	Agent-Based Model
What cognitive processes generate behaviour?	ACT-R

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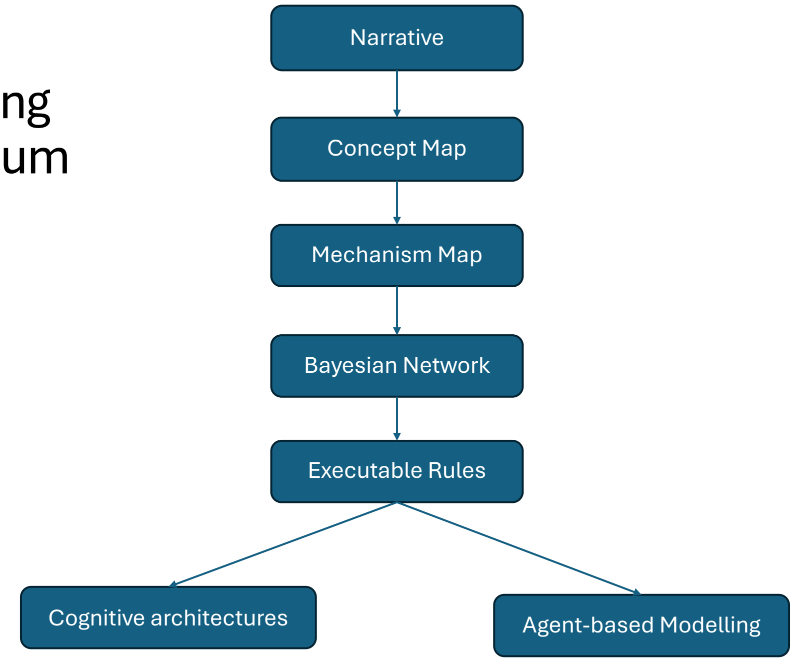
The modelling continuum



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Back to theory

- I introduced external representations of "theory" because they improve scientific reasoning:
  - External representations expand scientific understanding.
- The explanatory power of the theory/model, and consequently the amount and type of understanding it can provide, amounts to the number and importance of correct inferences it enables.

Synthese (2015) 192:3817–3837
DOI 10.1007/s11229-014-0591-2
CrossMark

External representations and scientific understanding

Jaakko Kuorikoski · Petri Ylikoski

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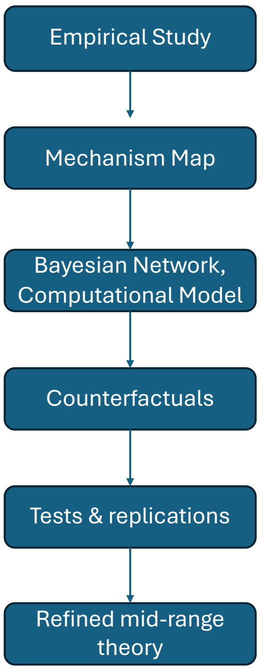
Cumulative inquiry



Scientific progress is cumulative when individual studies contribute to increasingly robust middle-range explanatory models.

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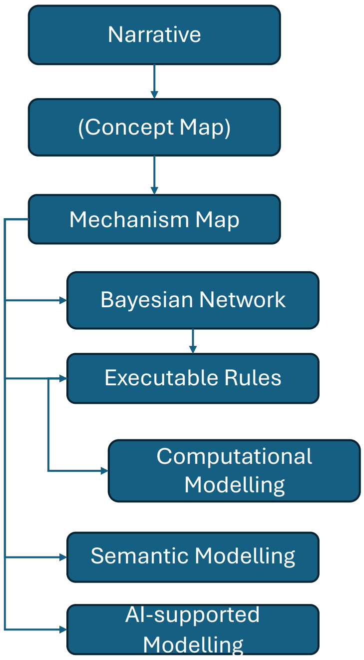

Conclusions

- Educational research does not progress only by collecting more data.
- It progresses by constructing better explanatory models.
- Those models are external representations that extend our ability to reason about learning mechanisms.

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General explanatory theories:
- Cognitive Load
- ACT-R
- Active Inference
- Distributed Cognition

Mechanism family:
Prediction
-> Comparison
-> Error detection
-> Model revision

Specific mechanisms:
AI generates feedback
-> student compares draft with feedback
-> student notices discrepancy
-> student revises argument

Mid-range theories contextualised

General theories act as "mechanism pump":

Active inference:
Predict
↓
Observe
↓
Compare
↓
Prediction error
↓
Update beliefs
↓
Act

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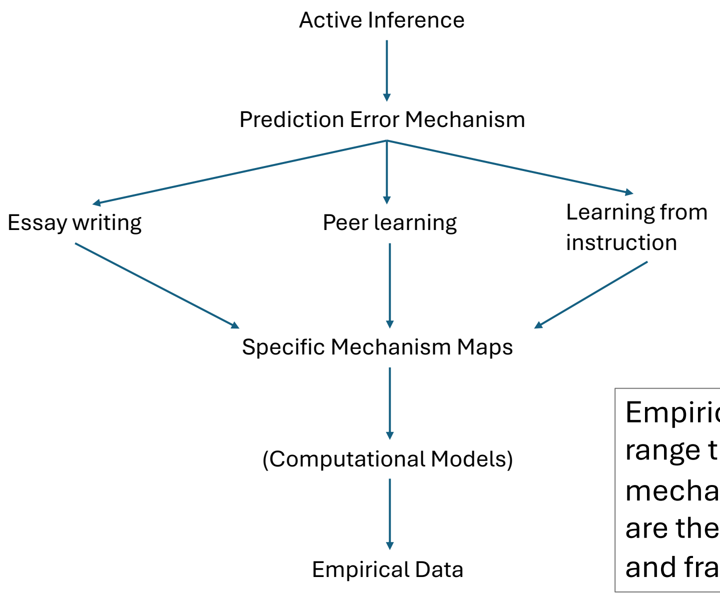

Empirical data speak to mid-range theories via specific mechanisms. Mid range theories are the bridge to general theories and frameworks.

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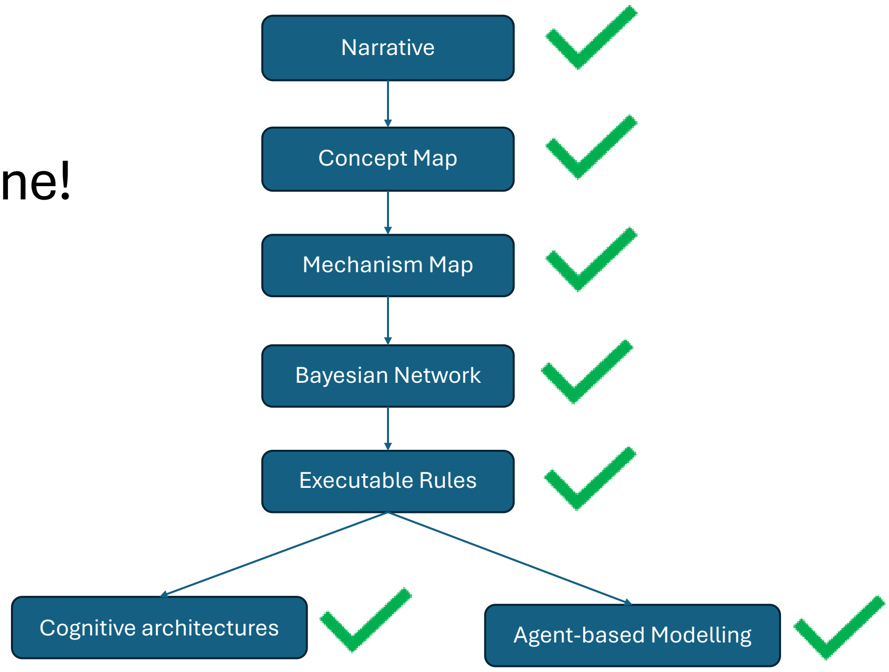
Well done!



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